

Meta Musicking

A Playground for Exploring Alternative Realities with Others in the XR Age

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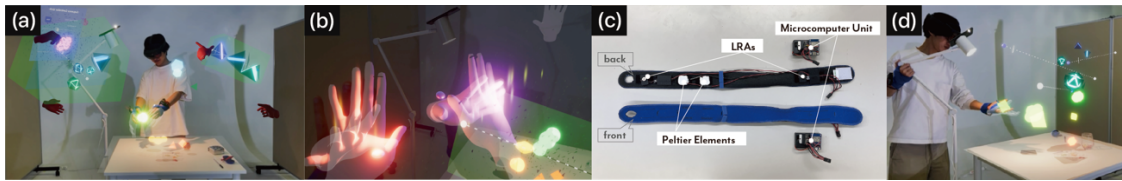


Figure 1: (a) A participant situated in real space is interacting with remote participants represented as hand avatars in a mixed reality environment. (b) By aligning the palms of the participants' hands, it is possible to generate a sustained or arpeggio sound with haptic feedback. (c) The flexible belt-type hot/cold haptic device. (d) Fusion of real shadow and virtual shadow with the XLS system.

CCS CONCEPTS

• Applied computing; • Arts and humanities; • Media Arts;

KEYWORDS

XR, MR, Haptics, Asymmetric Real-Space-Based Metaverse

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1 INTRODUCTION

Since 2017, we have been working on the "Drawing Sound in MR Space (DSMR)" project, aiming to develop new audiovisual art experiences in MR (Mixed Reality) space. Through this endeavor, we have recognized the significant role of "light and shadow" and "touch" in the XR (Cross Reality) experience. In our recent work "Meta Flowers [Sonobe et al., 2022]," referencing the research of Naemura et al. [Naemura et al., 2002], we investigated and implemented a system that combines real light with virtual shadows using a projector and a motion capture system. We refer to this

system as X Light-Shadow (XLS). While constructing XLS, we believe it's crucial to consider the implementation methods as part of the artwork experience. Additionally, in the work, we explored and implemented a glove-type haptic device equipped with linear resonant actuators (LRAs).

2 MOTIVATION

In addition to the aforementioned context, we also considered the state of the coronavirus pandemic, which led us to start contemplating the new DSMR experience. This experience intensifies engagement with the XR experience through touch, light, and shadow while remotely interacting with other participants. On the other hand, the term "Musicking" used in the project name is a concept proposed by Christopher Small, which perceives music not only as a performance but also as an activity [Small, 2012]. What would happen if we extended the musical experience, born from the collaboration of people and said to exist in any culture in the world, with our XR methods? This is the motivation and research question for this work, "Meta Musicking."

3 OVERVIEW OF META MUSICKING

In Meta Musicking, a multiparticipant remote MR experience, 1) each participant can interact with the hand avatars of other remote participants and virtual objects for musical experience within the physical space where each participant exists (see Figure 1, Figure 2), and 2) by touching their own index fingers together, or touching their index finger with that of other participants, they can create note objects that, when tapped, generate sound, visual effects, and

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Figure 2: (a) The conceptual drawing of Meta Musicking. Participants can experience it from any real space. (b) An installation set for exhibition and videography.

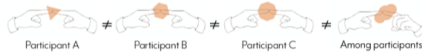


Figure 3: Note object generation: The shape and timbre of note objects generated by each participant will vary, as will the note objects that can be generated among participants.

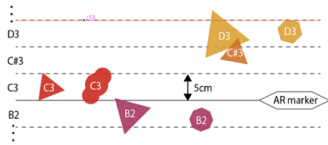


Figure 4: Pitch and color of note objects: The pitch of the note objects increases or decreases by a semitone for every 5 cm they are based the AR marker, and their color also changes.

haptic feedback with the belt-type hot/cold haptic device (see Figure 1 (c), Figure 3, Figure 4), and 3) they are able to freely compose and perform with these objects in mid-air (see Figure 1 (a)). Additionally, 4) it is possible to generate a sustained sound (drone sound) or arpeggio sound by aligning the palms of the participants' hands, and then 5) it is possible to feel the warmth and vibrations as the presence of remote others through haptic feedback with the haptic device (see Figure 1 (b) (c)). In addition, participants can see cross-shadow fusion between real and virtual shadows with the XLS of the artwork. Overall, we believe that through this musical play, each participant will not only be exposed to the possibilities of experiences beyond music but will also contemplate their relationship with others through remote communication via XR. Participants experience this artwork while wearing the HoloLens2 and the flexible belt-type hot/cold haptic device. Sharing of spatial coordinates is implemented using Photon Unity Network 2 (PUN2) and AR marker with Vuforia Engine. Additionally, iPhone12 is used as a position tracker to realize XLS of the artwork.

4 IMPLEMENTATION

4.1 Flexible Belt-type Hot/Cold Haptic Device

We have created a lightweight, wearable haptic device using LRAs and Peltier elements (see Figure 1 (c)). While a device composed of the same elements has been presented by Bentaleb et al. [Bentaleb et al., 2020], our device is based on a stretchable belt, allowing the user to arrange how it wraps around the hand-arm or head, and also to move and replace the two LRAs and two Peltier elements. The introduction of these flexible materials and movable components are intended to allow the flexible exploration of "sensory substitution" while experiencing them. A sensory substitution to VR experiences is being considered by Pezent et al. [Pezent et al., 2020].

4.2 X Light-Shadow of the artwork

By attaching an iPhone12, which can track the marker, to the light source position on the light stand, we synchronized the real light with the virtual light. With HoloLens2, the system rendered the virtual light, virtual shadows of the virtual objects, and combined virtual and real light-shadow to represent XLS (see Figure 1 (d)). This system operates on a local network and is realized by the light stand and iPhone12 at each participant's location.

4.3 Sequencer Unit

Participants can create a sequencer unit by tapping a button on the back of their left hand. The sequencer unit consists of starting-point and ending-point objects, and a plane object that repeatedly moves along the line connecting these two points from the starting to the ending point. When the plane object hits a note object, the sound and visual effects of the note object are generated. Additionally, participants can open the sequencer control unit and use the slider on the unit to change the size and speed of the plane object, meaning they can control the minimal elements necessary to create a piece of music. Participants can set up multiple sequencer units, making it possible to construct complex sound installations.

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